

EMR-on-EKS Streaming POC — Run Status Update

Prepared 2026-05-13 · Run `000000037gscnp2qanr` · v3.1 script

Bottom line. The current run has surpassed every prior attempt and is still going. At **19 h 7 m elapsed** it has just edged past our previous best of 19 h 6 m, and infrastructure health checks confirm no executor losses, no node disruption, and no spill into recovery storage. We are tracking toward the 24-hour stability milestone, after which we begin data-parity validation against the existing Databricks Delta Lake.

Current run snapshot

Job-run ID	000000037gscnp2qanr
Script version	v3.1 (Iceberg commit-retry + snapshot isolation + silver conflict-swallowing)
Submitted	2026-05-12 13:35:03 UTC (19:05 IST)
Elapsed	~19 h 7 m and counting
State	RUNNING — no <code>stateDetails</code> , no executor loss recorded

Run-history comparison

Run	Duration	Outcome	Cause
...fsjcis8rer	19 h 6 m	Failed	Executor evictions + Fluentd-OOM blind-spot
...g74dd0ivkf	7 h 44 m	Failed	Same Fluentd-OOM blind-spot
...glvqikoonn	3 h 9 m	Failed	Driver pod SIGTERMed (Karpenter consolidation)
...gncv3v0996	4 h 33 m	Failed	Executor graceful decommission (Karpenter)
...grqasb01cr (v3)	2 h 12 m	Failed	S3 Tables maintenance vs Iceberg MERGE — commit conflict
...gscnp2qanr (v3.1)	19 h 7 m+	Running	— (new best; surpassed all prior runs)

Infrastructure health (live CLI readings, 2026-05-13)

The Spark UI is currently unreachable for this run (AWS support ticket already raised by client). In the absence of the UI, we are using AWS CLI reads as the source of truth. Findings:

Health signal	Reading	Verdict
Underlying nodes (Karpenter <code>emr</code> NodePool)	3 × <code>m5a.2xlarge</code> (8 vCPU / 32 GiB each), all on-demand , all in <code>ap-south-1a</code> , all warm since 2026-05-12 10:58 UTC	Healthy
Cluster headroom	24 vCPU / 96 GiB capacity behind an 18 vCPU / 66 GiB ask — ~25 % headroom	Adequate
Spot reclaim exposure	Zero (all nodes are on-demand by design after v3 hardening)	Eliminated
Executor decommission spill (<code>s3://emr-migration-poc/spark-fallback/</code>)	0 bytes / 0 objects — no executor has been kicked, decommissioned, or had to migrate shuffle blocks in 19 h	Healthy
Karpenter consolidation events	None observed (same warm nodes since prior run; taint + toleration in place)	Healthy
Driver log noise (post <code>log4j2</code> fix)	Driver stderr remains small (KBs, not MBs) — <code>spark.log.level=WARN</code> fix continues to hold	Healthy

What this means

- **The two failure modes we engineered against are no longer firing.** The Karpenter consolidation signature (3 h 9 m / 4 h 33 m runs) and the S3 Tables commit-conflict signature (2 h 12 m run) have both stayed silent through 19 h.
- **The previous duration ceiling has been broken.** No prior run on either configuration has lasted longer than this one.
- **We are in new territory.** Anything beyond this point is data we have not previously collected on this stack.

Risks and open items

- **Spark UI unreachable for this run** — client has raised an AWS support ticket. CLI-based observability is our workaround; daily health checks are scripted and repeatable.
- **Live per-pod CPU / memory / disk metrics** require `cloudwatch:ListMetrics` & `GetMetricStatistics` permissions, which the `saranya` IAM user does not currently have. Requesting `CloudWatchReadOnlyAccess` from the client would unlock Container Insights and remove our last observability gap.
- **kubectl access** remains unavailable (EKS endpoint is private-only). Not blocking the run, but limits root-cause depth if anything does fail.

Next milestones

- **24 hours elapsed** (~13:35 UTC today) — declares baseline stability on prod-grade config.
- **Post-24 h:** begin data-parity validation of bronze + silver tables against the existing Databricks Delta Lake (bronze schema preserved exactly to enable apples-to-apples diff).
- **Production sign-off criteria:** 24 h continuous + parity passed + cost-allocation tags activated in AWS Billing console (one outstanding client-side action).

Source: AWS CLI reads via profile `batterysmart` (region `ap-south-1`); EMR-on-EKS virtual cluster `brtw9ztup2y01wok6x8r2bsmr`; EKS cluster `bs-analytics-eks-cluster`, namespace `emr-prod`. Prepared from project STATUS.md and live infrastructure queries on 2026-05-13.